School of Computing and Mathematical Sciences

CO7201 Individual Project

Preliminary Report

Coloring Black Boxes - Visualization of Decision Making of AI

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1. Introduction

Satellite imagery is central to how we monitor land use, track environmental change, and plan infrastructure. Getting the classification of that imagery wrong has real consequences — a misidentified land cover type can skew a policy decision or delay a response. Deep learning has made classification more accurate, but it has also made the reasoning behind each decision harder to see. That opacity is the starting point for this project.

Explainable AI tools — particularly those that produce visual heatmaps over an image — are one way to address this. They highlight the regions a model found relevant to its prediction. The problem is that almost all research evaluating these tools has focused on convolutional networks. Three very different architectural types are now widely used: CNNs, which build features locally through filters; Vision Transformers, which consider relationships across the entire image at once; and Mamba, a new architecture that processes spatial information through a kind of running compressed memory. Each works differently, and there is no reason to assume that explanation methods will behave the same way across all three.

This project will train one model from each family on the EuroSAT satellite dataset and compare how explanation quality holds up across them. The research question is:

> ***How does the choice of neural network architecture affect the faithfulness, stability, sensitivity, and diagnostic usefulness of visual explanations in satellite image classification?***

2. Project Aims and Objectives

2.1 Aim

To systematically compare visual XAI methods across CNN, Vision Transformer, and Vision Mamba models trained on EuroSAT, using quantitative metrics to evaluate how architecture influences explanation quality.

2.2 Research Question

How does neural network architecture (CNN, ViT, Mamba) shape the faithfulness, stability, sensitivity, and diagnostic value of visual explanations in satellite image classification?

2.3 Objectives

Essential — must be completed:

- Review literature on CNNs, ViTs, Mamba/SSMs, XAI methods, and explanation evaluation.
- Prepare EuroSAT: stratified 70/15/15 split, same seed across all models.

- Train ResNet-18, ViT-Tiny, and Vision Mamba; record accuracy, F1, and confusion matrices.
- Implement Integrated Gradients and SmoothGrad via Captum for all three architectures.
- Implement Grad-CAM/Grad-CAM++ (CNN) and Attention Rollout (ViT) as architecture-specific tools.
- Build evaluation pipeline: faithfulness (insertion/deletion AUC), stability (cosine similarity), sensitivity (occlusion drop).
- Run failure-case analysis to find architectural biases and shortcut-learning patterns.
- Produce a cross-architecture comparison with quantitative tables and annotated visuals.

Desirable — if time allows:

- Compare two variants in one family (e.g. ResNet-18 vs ResNet-50).
- Add LIME or SHAP for a model-agnostic cross-method baseline.
- Class-level explanation breakdown per EuroSAT category.
- Interactive Colab notebook for predictions and explanations.

Optional — aspirational:

- Small user study on explanation usefulness (ethics approval needed).
- t-SNE/UMAP of learned representations per architecture.
- Replicate evaluation on UC Merced dataset.

3. Background and Related Work

3.1 CNNs in Vision

CNNs detect image features through filters that scan small local regions, stacking layers to build progressively abstract representations. The key detail for this project is that this process leaves behind spatial maps — grids that record where in the image each feature was detected. Those maps are precisely what gradient-based XAI methods like Grad-CAM depend on. Without them, Grad-CAM cannot work. ResNets (He et al., 2016) made very deep convolutional networks trainable and became the standard baseline across vision tasks. EuroSAT (Helber et al., 2019) was built to benchmark deep learning classifiers on satellite imagery, and CNNs were the primary model evaluated there.

3.2 Vision Transformers

Rather than scanning local windows, a Vision Transformer (Dosovitskiy et al., 2020) splits an image into patches and asks each patch to weigh its relationship with every other patch simultaneously. This global view is powerful — it can link two distant parts of an image that a CNN would need many layers to connect. But it creates an immediate problem for XAI: there are no spatial feature maps in the CNN sense, so methods that rely on them do not transfer. On a 64×64 image with patch size 16, the model processes just 16 patches, which keeps computation manageable.

3.3 Vision Mamba

Mamba (Gu and Dao, 2023) processes sequences by maintaining a compressed running state that is updated at each step, rather than comparing all elements simultaneously. VMamba (Liu et al., 2024) and Vim (Zhu et al., 2024) apply this to images by scanning patches in spatial order. It is faster than a transformer for long sequences, but from an XAI perspective it is almost entirely unexplored — no standard explanation method has been developed for it, and no pretrained ImageNet weights are available, meaning it must be trained from scratch in this project.

3.4 Explainability Methods

Grad-CAM (Selvaraju et al., 2017) identifies spatially important regions by examining which CNN feature maps contributed most to a particular prediction. It is specific to convolutional models. Integrated Gradients (Sundararajan et al., 2017) works differently and more broadly: it measures each pixel's contribution by tracking how the model's output shifts as the input is gradually built up from a blank baseline. Because it asks a question about the input rather than the architecture's internals, it works on any model — making it the central comparison tool across all three architectures here. SmoothGrad (Smilkov et al., 2017) reduces the visual noise in gradient maps by averaging results across many slightly altered versions of the same input. Attention Rollout (Abnar and Zuidema, 2020) tracks where a transformer distributes its attention across layers. It is included as a visual reference only — research by Jain and Wallace (2019) and Wiegreffe and Pinter (2019) has shown that where a model attends and what actually drove its output are not the same thing.

3.5 Evaluation of Explanations

Three quantitative measures will be used, drawn from the XAI evaluation literature. Petsiuk et al. (2018) introduced insertion and deletion tests: if an explanation correctly identifies what the model relied on, then uncovering those regions first should quickly restore prediction confidence, and hiding them first should quickly destroy it. The area under each curve gives a faithfulness score. Alvarez-Melis and Jaakkola (2018) proposed testing stability: a good

explanation should not change much when a small imperceptible perturbation is added to the image. Yeh et al. (2019) introduced sensitivity: if an explanation marks a region as important, hiding that region should cause a noticeable drop in confidence. Together these three allow explanation quality to be measured and compared, rather than just inspected.

4. Technical Specification

4.1 Dataset

EuroSAT (Helber et al., 2019): 27,000 Sentinel-2 satellite images, 10 land-use classes, 64×64 pixels. RGB bands only (for ImageNet weight compatibility). Split: 70% train, 15% validation, 15% test — stratified, seed fixed at 42, identical across all models. Standard augmentations applied to training set only.

4.2 Models

- **ResNet-18:** ImageNet pretrained via torchvision; classification head replaced; AdamW (lr=1e-4), cosine schedule, 30 epochs.
- **ViT-Tiny:** ImageNet-21k pretrained via timm; patch size 16 (16 tokens on 64×64); same training protocol.
- **Vision Mamba (Vim/VMamba):** Trained from scratch; 5-epoch warmup; AdamW + cosine decay; FP16 mixed precision. If unstable, lr reduced to 1e-5.

4.3 XAI Methods

- **Integrated Gradients:** Captum IntegratedGradients; zero-tensor baseline; 50 steps. All three models.
- **SmoothGrad:** Captum NoiseTunnel over IG; 50 samples; $\sigma = 0.1 \times$ input range. All three models.
- **Grad-CAM / Grad-CAM++:** Captum LayerGradCam on ResNet layer4; upsampled to 64×64. CNN only.
- **Attention Rollout:** All-layer ViT attention propagation per Abnar and Zuidema (2020). Qualitative reference only.

4.4 Tools and Libraries

- **PyTorch 2.x, torchvision, timm:** Training framework and pretrained weight loading. Autograd required for all gradient methods.
- **Captum:** Tested XAI library — avoids custom gradient implementation errors.
- **scikit-learn, NumPy, SciPy:** Metrics, stratified splits, statistical tests.
- **OpenCV, matplotlib:** Heatmap overlays, occlusion masks, evaluation plots.
- **W&B / MLflow:** Experiment logging; all seeds fixed to 42.

- **Hardware:** Google Colab Pro (T4/A100); university HPC as fallback.

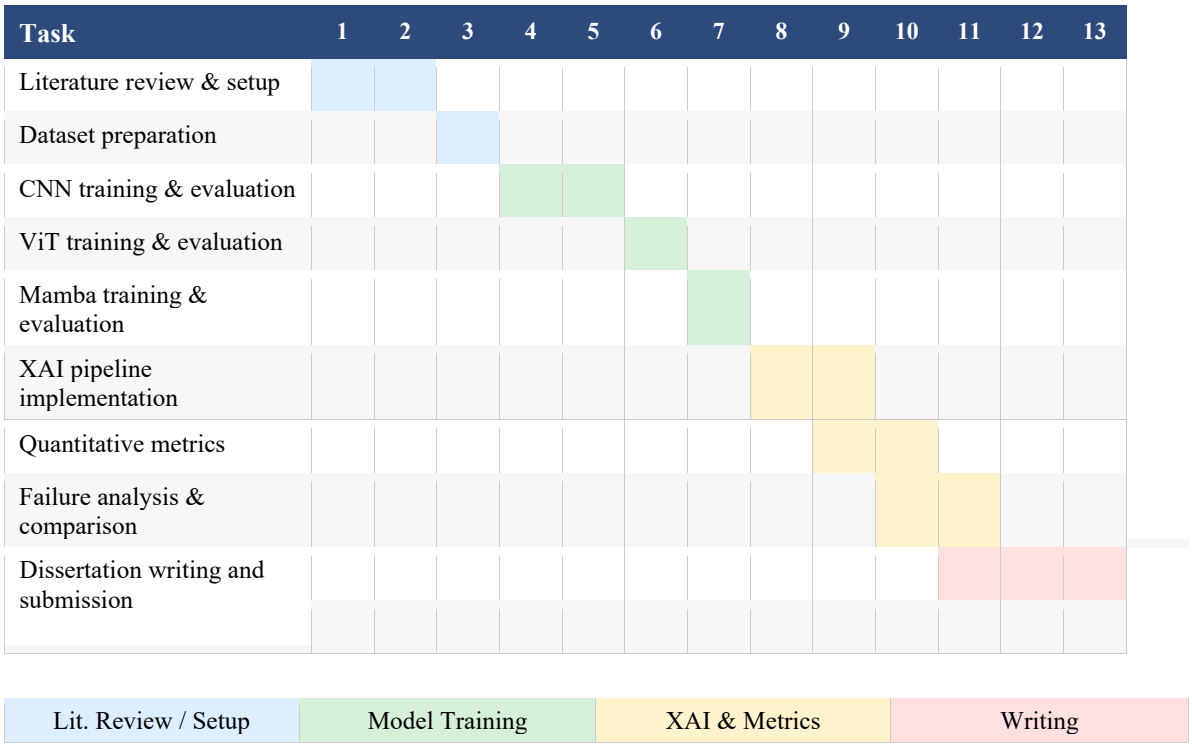
5. Requirements Evaluation Plan

Each objective is tied to a measurable method and a clear pass criterion.

Metric	Method	Pass Criterion	Links to Objective
Faithfulness	Insertion & deletion AUC on 200 test images per model	Insertion AUC > Deletion AUC for all models	Objectives 4, 5, 6
Stability	Cosine similarity of explanation under σ=0.01/0.05/0.1 noise	Mean cosine sim ≥ 0.80 at σ=0.01	Objective 6
Sensitivity	Probability drop after occluding top-20% attribution region	Average drop ≥ 10%	Objective 6
Qualitative	Annotated failure cases per model	≥3 bias patterns per model	Objectives 7, 8

6. Time Plan and Milestones

All three models are trained before the XAI pipeline starts (Week 8) so the evaluation framework can be reused consistently. Writing runs in parallel from Week 11.



Milestones: M1 (W2) Literature done | M2 (W5) CNN baseline | M3 (W6) ViT baseline | M4 (W7) Mamba baseline | M5 (W9) XAI pipeline complete | M6 (W10) All metrics computed | M7 (W11) Failure analysis done | M8 (W13) Submitted

7. Risk Analysis

Risk	Likelihood	Impact	Mitigation
Mamba has no pretrained weights and may be unstable	Medium	High	Train with warmup LR; use VMamba or Vim; fall back to simpler SSM if needed
Colab GPU quota may run out	Medium	Medium	Use FP16, small models, frequent checkpoints; university HPC as backup
Insertion/deletion tests are slow at scale	Low	Medium	Batch processing; cap at 20 steps; profile before full run
No standard XAI method exists for Mamba	Low	Medium	Use model-agnostic IG as primary; attempt gradient attribution; document gaps
Running out of time on optional objectives	Medium	High	Finish all Essential work first; Desirable/Optional only if time genuinely allows

8. Expected Outcomes

1. Three trained, documented models on EuroSAT with reproducible code.
2. A working XAI pipeline producing IG, SmoothGrad, Grad-CAM, and Attention Rollout maps.
3. Quantitative evaluation results (faithfulness, stability, sensitivity) in comparative tables.
4. Failure-case analysis identifying architectural bias patterns across all three models.
5. A critical discussion of how architecture shapes explanation behaviour, with practical recommendations.
6. A complete MSc dissertation.

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